$$
\begin{aligned}
70 &= k(25+5-100) \\
\Rightarrow k &= -1 \\
\text{Hence } x &= 100 - p - p^2 \\
R &= px \\
\text{Revenue} &= p(100 - p - p2)
\end{aligned}
$$

Exercise 3.2

1. The cost of over haul of an engine is ₹10,000 The operating cost per hour is at the rate of $2x-240$ where the engine has run x km. Find out the total cost if the engine run for 300 hours after overhaul.

2. Elasticity of a function $\frac{Ey}{Ex}$ is given by $\frac{Ey}{Ex} = \frac{-7x}{(1-2x)(2+3x)}$. Find the function when $x = 2, y = \frac{3}{8}$.

3. The elasticity of demand with respect to price for a commodity is given by $\frac{(4-x)}{x}$, where p is the price when demand is x. Find the demand function when price is 4 and the demand is 2. Also find the revenue function.

4. A company receives a shipment of 500 scooters every 30 days. From experience it is known that the inventory on hand is related to the number of days x. Since the shipment, $I(x) = 500 - 0.03x^2$, the daily holding cost per scooter is ₹ 0.3. Determine the total cost for maintaining inventory for 30 days.

5. An account fetches interest at the rate of 5% per annum compounded continuously An individual deposits ₹1,000 each year in his account. How much will be in the account after 5 years. $(e^{0.25} = 1.284)$.

6. The marginal cost function of a product is given by $\frac{dC}{dx} = 100 - 10x + 0.1x^2$ where x is the output. Obtain the total and the average cost function of the firm under the assumption, that its fixed cost is ₹ 500.

7. The marginal cost function is $MC = 300\ x^{\frac{2}{5}}$ and fixed cost is zero. Find out the total cost and average cost functions.

8. If the marginal cost function of x units of output is $\frac{a}{\sqrt{ax+b}}$ and if the cost of output is zero. Find the total cost as a function of x.

9. Determine the cost of producing 200 air conditioners if the marginal cost (is per unit) is $C'(x) = \frac{x^2}{200} + 4$.

10. The marginal revenue (in thousands of Rupees) functions for a particular commodity is $5 + 3\ e^{-0.03x}$ where x denotes the number of units sold. Determine the total revenue from the sale of 100 units. (Given $e^{-3} = 0.05$ approximately)

11. If the marginal revenue function for a commodity is $MR = 9 - 4x^2$. Find the demand function.

12. Given the marginal revenue function $\frac{4}{(2x+3)^2} - 1$, show that the average revenue function is $P = \frac{4}{6x+9} - 1$.

13. A firm's marginal revenue function is $MR = 20\,e^{-x/10}\left(1 - \frac{x}{10}\right)$. Find the corresponding demand function.

14. The marginal cost of production of a firm is given by $C'(x) = 5 + 0.13x$, the marginal revenue is given by $R'(x) = 18$ and the fixed cost is ₹ 120. Find the profit function.

15. If the marginal revenue function is $R'(x) = 1500 - 4x - 3x^2$. Find the revenue function and average revenue function.

16. Find the revenue function and the demand function if the marginal revenue for x units is $MR = 10 + 3x - x^2$.

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